

Visual perception involves the organization and interpretation of information obtained from visible light. Visual perception relies on monocular and binocular cues, as well as *perceptual organization principles* that allow humans to interpret limited or partial information more holistically. These sorts of mental shortcuts are based on prior experiences and knowledge, and are especially useful when lighting or perspective is limited. For example, the ambiguous image in Figure 1 is often perceived as a cup.



Figure 1 Ambiguous image

Perceptual organization principles allow our brains to quickly and efficiently make assumptions about visual inputs, but these principles can also give rise to perceptual illusions. For example, the Ames room is constructed to exploit monocular depth cues in such a way that the person standing on one side of the room appears to impossibly dwarf the person on the other side of the room (Figure 2).

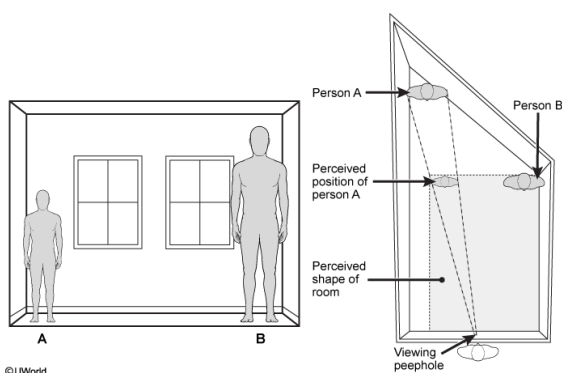


Figure 2 Ames room illusion (two people standing in a specially constructed Ames room, viewed through a monocular peephole)

Human senses were adapted for use on land, which can lead to potentially deadly perceptual distortions while flying or underwater diving. Pilots are prone to such perceptual distortions when landing on unfamiliar runways. During their training, pilots develop a mental image of how a typical runway appears as they descend and land. Later, they

compare this prototypical shape to the runway they see before them to make adjustments

in the slope of their descent. However, if a runway differs from what the pilot is used to (eg, in width or slope), it will differ from the pilot's mental image for a typical runway at that same altitude. For example, an unusually wide runway will appear to be closer than it actually is, and a narrow runway will seem to be further away. This can cause the pilot to approach at a dangerously steep or shallow slope.

Pilots and divers can also suffer from spatial disorientation due to variations in sensory stimuli that are not typically encountered on land. For example, gravity is not experienced in the body the same way underwater as on land, and visual cues are often limited in deep dives. In these murky conditions, divers can easily lose track of the surface of the water, leading to extreme spatial disorientation and panic.

Which of the following predictions is best supported by information presented in the passage?

- ☐ A. Individuals will perceive a cup in Figure 1 despite differences in expectations. [13%]
- ☐ B. Visual principles of perceptual organization help prevent optical illusions from occurring. [7%]
- ☐ C. According to the law of continuity, spatial disorientation will be greater under water than in the air. [9%]
- ✓ ☒ D. The Ames room configuration will override the perceptual principle of size constancy. [68%]

Correct

69% answered correctly

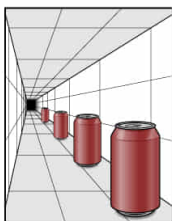
Explanation:



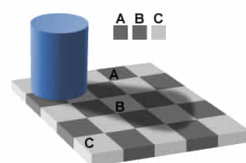
Shape constancy
Shape of door appears to be the same whether opened or closed



Color constancy
Colors of fruit appear to remain the same under different lighting conditions



Size constancy
Distant object appears to be the same size as near object



Brightness constancy
Block B appears to be same brightness as C (really matches A)

Constancy refers to our perception that an object's **characteristics remain stable** even when the sensory input from that object changes due to alterations in the environment such

as distance or lighting. Constancy applies to **shape**, **color**, **brightness**, and **size**. For example, the shape of a door is perceived as a rectangle whether it is open or closed; a

banana is seen as yellow despite shadow or lighting; and with proper depth and distance cues, an object's size appears to stay constant even when the distance increases (and the retinal image gets smaller).

The Ames room is constructed so that the principle of size constancy is overridden. Because the viewer can see the room with only one eye (through a monocular peep hole), binocular depth cues are absent and the arrangement of the room is in a trapezoidal configuration. However, to the viewer, it appears that the room is constructed as a normal rectangle and that person A is the same distance away and much smaller than person B.

(Choice A) Principles of perceptual organization are top-down cognitive processes, which are guided by prior experience and expectations. Therefore, differences in expectations might cause one person to perceive a cup while another perceives a wastebasket, a thimble, or any number of other objects.

(Choice B) Rather than visual principles of perceptual organization helping to *prevent* optical illusions from occurring, they actually explain in part why optical illusions do occur. These principles allow our brains to use mental shortcuts to process visual information more quickly, which can also give rise to perceptual illusions.

(Choice C) The law of continuity is a Gestalt principle of perceptual organization that suggests that we tend to perceive objects that are in the same row, or lined up, as together. The law of continuity does not predict that spatial disorientation will be greater under water than in the air.

Educational objective:

Constancy refers to our perception that characteristics of an object (eg, shape, color, brightness, size) remain stable even when the sensory input from that object changes due to alterations in the environment (eg, distance, lighting).

Time spent:0 sec

Subject:

Behavioral Sciences

QID:400279

System:

Sensation, Perception, and Consciousness

Visual perception involves the organization and interpretation of information obtained from visible light. Visual perception relies on monocular and binocular cues, as well as *perceptual organization principles* that allow humans to interpret limited or partial information more holistically. These sorts of mental shortcuts are based on prior experiences and knowledge, and are especially useful when lighting or perspective is limited. For example, the ambiguous image in Figure 1 is often perceived as a cup.



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Figure 1 Ambiguous image

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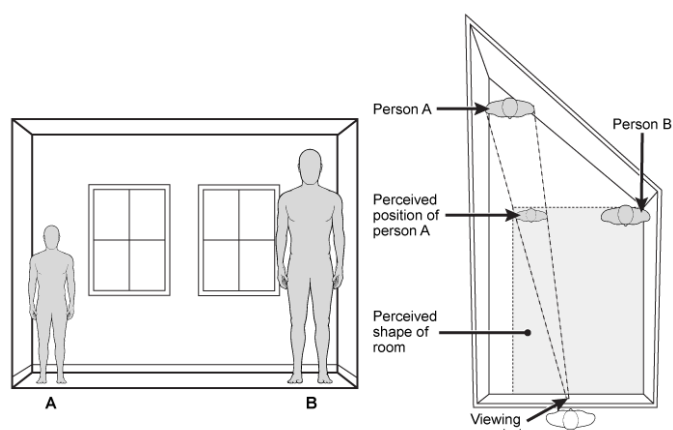


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prone to such perceptual distortions when landing on unfamiliar runways. During their training, pilots develop a mental image of how a typical runway appears as they descend and land. Later, they compare this prototypical shape to the runway they see before them to make adjustments in the slope of their descent. However, if a runway differs from what the pilot is used to (eg, in width or slope), it will differ from the pilot's mental image for a typical runway at that same altitude. For example, an unusually wide runway will appear to be closer than it actually is, and a narrow runway will seem to be further away. This can cause the pilot to approach at a dangerously steep or shallow slope.

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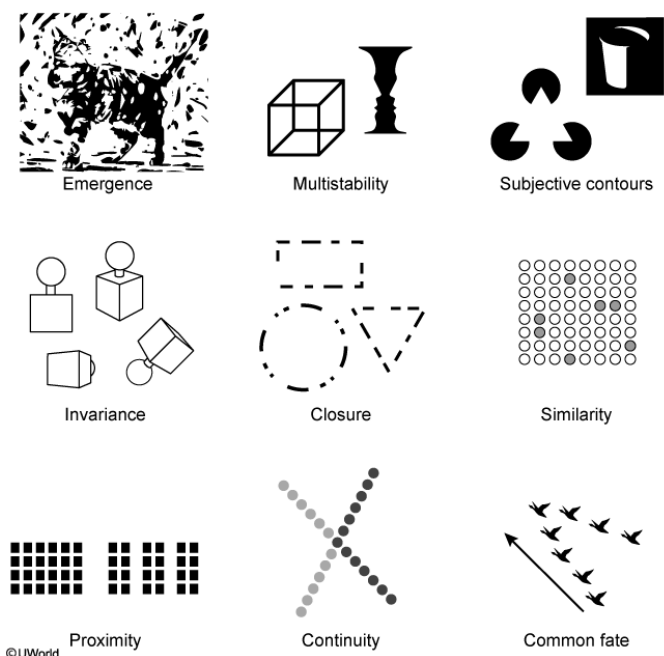
The perception of a cup in Figure 1 is best attributed to which Gestalt principle of perceptual organization?

- ☐ A. Invariance [3%]
- ☐ B. Common fate [9%]
- ☐ C. Similarity [18%]
- ✓ ☒ D. Subjective contours [68%]

Correct

69% answered correctly

Explanation:



The **Gestalt principles** of perceptual organization describe how humans **holistically perceive** sensory stimuli. Although Gestalt principles can apply to any sensory stimulus, they are most often described using visual stimuli. Many Gestalt principles aid in shaping our visual perception to help us make sense of what is seen.

For instance, the principle of **subjective contours**, also known as **reification** or illusory contours, describes how our mind fills in the gaps. In Figure 1, the subjective contours of a circle or a cup are perceived despite their **contours not being explicitly shown**.

(Choice A) Invariance suggests that objects can be recognized despite alterations in orientation, lighting, scale, and slight alterations in the objects' component features. This allows for the recognition of an object in different spatial orientations or when it is depicted with dotted rather than solid lines.

(Choice B) Common fate is a principle that describes our perception that objects that move together are grouped together.

(Choice C) Similarity is the principle that objects sharing certain features (such as shape or color) are grouped together.

Educational objective:

Gestalt principles of perceptual organization describe how holistic perception of sensory stimuli occurs. One example is the subjective contours principle, which allows us to "fill in" contours that are not explicitly shown.